XYZ File Formats

VSEPR-Sim Atomistic Engine — Formats: `.xyz` `.xyza` `.xyzc` `.xyzf`

1 Overview

All VSEPR-Sim molecular I/O uses plain-text, line-oriented files. Four formats share a common structural grammar; each adds fields to the preceding one.

Ext.	Name	Use case
`.xyz`	Standard XYZ	Static geometry snapshot
`.xyza`	Extended XYZ	Charges, velocities, forces
`.xyzc`	Checkpoint XYZ	Restartable MD state
`.xyzf`	Formation XYZ	Multi-frame trajectory

Units (all formats): coordinates in Å (10^{-10} m); energies in kcal/mol; velocities in Å/fs; forces in kcal/(mol Å); charges in elementary charge e.

2 Standard XYZ (`.xyz`)

2.1 Grammar

```
<N>
<comment>
<sym>  <x>  <y>  <z>
...   (N lines)
```

- **Line 1**: atom count N (integer ≥ 1)
- **Line 2**: free-text comment (may be empty)
- **Lines 3–**: element symbol + three Cartesian coordinates, whitespace-separated

2.2 Coordinate system

Right-handed Cartesian (x, y, z) in Å. Origin is arbitrary; conventional choice is near the centre of mass. Precision: 6 decimal places $\Rightarrow$ resolution ≈ 0.1 pm.

2.3 Example: water

```
3
H2O | E = -6.245 kcal/mol
O   0.000000  0.000000  0.119262
H   0.000000  0.763239 -0.477049
H   0.000000 -0.763239 -0.477049
```

2.4 Comment line convention

```
<name> | E = <val> kcal/mol | <notes>
```

3 Extended XYZ (`.xyza`)

Adds per-atom scalar and vector properties after the (x, y, z) fields.

3.1 Grammar

```
<N>
<comment> properties="<list>"
<sym> <x> <y> <z> [<q> <vx> <vy> <vz> <fx>
     <fy> <fz> <e>]
```

The `properties` key in the comment line declares which optional columns are present (order is fixed):

Key	Symbol	Unit	Cols
`charge`	q_i	e	1
`velocity`	$\mathbf{v}_i$	Å/fs	3
`force`	$\mathbf{F}_i$	kcal/(mol Å)	3
`energy`	ε_i	kcal/mol	1

3.2 Example: H_2O with charges and velocities

```
3
Water MD frame 1 properties="charge:
    velocity"
O  0.00  0.00  0.12 -0.834  0.00  0.00
    0.00
H  0.00  0.76 -0.48  0.417  0.12  0.08
    -0.03
H  0.00 -0.76 -0.48  0.417 -0.12  0.08
    -0.03
```

4 Checkpoint XYZ (`.xyzc`)

Extends `.xyza` with a simulation-state header block required to restart an interrupted MD run without loss of trajectory continuity.

4.1 Header block (before atom data)

```
CHECKPOINT
step    <integer>
time    <float> fs
dt      <float> fs
T_target <float> K
seed    <integer>
box     <ax> <ay> <az>  (Angstrom,
    optional PBC)
END_CHECKPOINT
<N>
<comment>
<atom lines with charge+velocity+force>
```

- `step`: MD step number at checkpoint
- `time`: accumulated simulation time (fs)
- `dt`: integrator timestep (fs)
- `T_target`: thermostat set-point (K); 0 = NVE
- `seed`: RNG seed for reproducible continuation
- `box`: simulation box vectors; omit for open boundary

5 Formation XYZ (`.xyzf`)

A multi-frame format: one or more standard `.xyz` or `.xyza` blocks concatenated in a single file, separated only by the next atom-count line. Used for minimisation traces, NVT trajectories, and reaction paths.

```
<N>          <- frame 1
<comment>
<atom lines>
<N>          <- frame 2
<comment>
<atom lines>
...
```

Each frame is self-contained and independently parseable. The comment line of frame k typically records the step index and instantaneous energy:

```
Step 0 | E = -12.30 kcal/mol | FIRE iter
Step 1 | E = -13.11 kcal/mol | FIRE iter
```

6 Bounding Box & PBC

When periodic boundary conditions are active, the box geometry is stored either in the `.xyzc` header or as a special comment-line key in any format:

```
Lattice="ax 0 0  0 ay 0  0 0 az" pbc="T T
    T"
```

Box vectors are in Å. `pbc` flags indicate which axes are periodic (T) or open (F). The format is compatible with the ASE *Extended XYZ* convention.

7 Implementation

File	Role
`include/io/xyz_format.hpp`	Struct definitions, API
`src/io/xyz_format.cpp`	Reader / writer impl.
`include/thermal/xyzc_format.hpp`	Checkpoint reader
`src/thermal/xyzc_format.cpp`	Checkpoint writer

Key reader types: `XYZReader`, `XYZAReader`, and `XYZWriter` / `XYZAWriter` in `namespace vsepr::io`. Bond auto-detection (`detect_bonds`) uses per-element covalent radii with a configurable scale factor (default 1.2).

8 Typical Energy Ranges

Observable	Typical range	Unit
Total bonded energy	$-10^2 - -10^5$	kcal/mol
Relative conformer	$0.1 - 10$	kcal/mol
Activation barrier	$5 - 50$	kcal/mol
Force (unrelaxed)	$1 - 100$	kcal/(mol Å)
Force convergence	< 0.01	kcal/(mol Å)
Velocity (300 K, C)	~ 0.02	Å/fs

See also: `docs/FILE_FORMATS.md` for prose description, `section5_integration.tex` for MD timestep conventions.